Cost, Carbon, and the Speed of Replenishment in Distribution Network Design

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Abstract

To distribute a product, a firm operates a network of distribution centers and replenishes them to serve its markets. How densely it places those centers and how fast it replenishes them shape both its cost and the carbon its distribution emits, and as carbon is increasingly priced and reported these choices are no longer settled by cost alone. We study how a carbon price sets the two levers in a model of a firm distributing a single product from one source to many markets, minimizing cost subject to a carbon cap and a service requirement, with carbon arising both in moving goods and in holding inventory. The familiar response — replenish more slowly and consolidate the network — lowers transport carbon but enlarges the inventory whose holding also emits, so whether it lowers *total* carbon depends on which activity is more carbon-intensive per dollar. We show that a carbon price can instead call for *faster* replenishment, when holding is the more carbon-intensive way to keep products available, and for a *more dispersed* network, when a long last mile is the more carbon-intensive way to stay close to markets; because the two levers are complements, it can call for both at once. Closed-form thresholds — a speed threshold and a pooling-adjusted density threshold — classify products by the operational substitute through which carbon is reduced. The analysis also shows that transport-only carbon measures may shift emissions into inventory and warehousing, whereas a total-logistics accounting boundary prices both sources and preserves the relevant speed–density trade-off.

1 Introduction

A firm serves its markets from a network of distribution centers that it replenishes over time. Among the operational choices this involves, we study two: the *density* of the network — how many distribution centers the firm operates — and the *speed* with which those centers are replenished. Both shape the firm's cost, and both shape the carbon its distribution emits. As carbon is increasingly priced and reported (McKinnon and Piecyk, 2012), a firm sets these levers with carbon in view alongside cost, and we study how a carbon price shapes that choice.

Each lever works through a standard cost–carbon tradeoff. Network density governs proximity to markets: operating more distribution centers shortens the last-mile distance and changes the

warehouse energy the network consumes. Replenishment speed governs availability: replenishing faster shortens the lead time that inventory must cover and changes the emissions of the freight that serves the network. Distribution emits carbon both when goods move and when they are held, and each lever affects both the firm's cost and its carbon. Cost balances these effects one way; with a price on carbon, the carbon-minimizing balance can fall elsewhere, and how it does so is the question we take up.

Slowing replenishment and consolidating the network are the standard transport-side levers: slowing moves freight by slower, lower-emission modes, the logic of slow steaming in shipping (Psaraftis and Kontovas, 2013; Cariou, 2011), and consolidating serves markets from fewer, larger distribution centers (McKinnon and Piecyk, 2012). Each response also moves inventory. A unit held in a warehouse consumes energy — refrigeration, climate control, handling — for as long as it is held. Slowing replenishment lengthens the lead time that a stock buffer must cover, so the safety stock, and the carbon of holding it, rises with the square root of that lead time; consolidating pools demand across markets (Eppen, 1979) and saves facility energy while lengthening the last mile. Each move thus lowers one source of carbon and raises another, and whether it lowers the total depends on which source is larger.

The choice between the two sources is, at root, a choice between two operational substitutions. A firm provides *availability* either by holding safety stock or by replenishing faster — a shorter lead time leaves less for the buffer to cover, so speed and inventory are substitutes for the same service — and it provides market *proximity* either by operating more local facilities or by delivering farther from fewer, so facilities and last-mile travel are substitutes for the same access. Cost selects one mix of these currencies; carbon, weighting the emissions of each, can select another. When safety stock is more carbon-intensive per dollar than the inbound transport that would replace it, carbon favors buying availability with speed — replenishing *faster*. When last-mile travel is more carbon-intensive per dollar than the facilities that would shorten it, carbon favors buying proximity with facilities — a *more dispersed* network. The lowest-carbon distribution system thus depends on the product: a cold-chain grocer and a bulk dry-goods shipper land in different configurations.

We make this precise in a model of a firm that distributes a single product from one source through a network of distribution centers to its markets under stochastic demand. The firm chooses how dense to make the network and how fast to replenish it, so as to minimize cost subject to a carbon cap and a service-level requirement. Cost and carbon each comprise the same four pieces — facilities, holding, inbound transport, and last-mile transport — so carbon splits cleanly into a transport bucket and a holding bucket. This is the classical location–inventory problem (Daskin et al., 2002; Shen et al., 2003) — here in a continuous-approximation form (Daganzo, 2005) rather than the combinatorial mixed-integer formulation of those papers — placed under a carbon constraint, with one addition that drives everything: the replenishment *speed* is a decision. Because speed sets the lead time that stochastic safety stock must cover, it is the firm's direct lever on inventory carbon, and it couples the two buckets.

The analysis is organized around the *marginal carbon intensity per dollar* of each activity and

yields three structural results and a map. First, a *speed reversal* in the availability substitution: holding the network fixed, total carbon is U-shaped in speed, and the carbon-minimizing speed exceeds the cost-minimizing one when holding is more carbon-intensive per dollar than inbound transport at the margin, so a carbon constraint can call for faster rather than slower replenishment — a distinct configuration from the slow-steaming view. Second, a *pooling-adjusted density threshold* in the proximity substitution. The carbon-minimizing network is more dispersed when the last mile is carbon-intensive per dollar relative to facilities. Adding a facility, however, also gives up some demand pooling, so the threshold is not that facility comparison alone: it blends in the holding intensity, and therefore depends on demand correlation, service, and volatility. Third, the two levers are *complements*: faster replenishment shrinks safety stock and lowers the inventory penalty of dispersing, so the carbon-minimizing response moves both together rather than one alone. Together the two thresholds sort any product into one of four regimes by the direction in which carbon shifts its speed and its network. The contribution is this marginal carbon-intensity logic for locating a product, not the *managerial map* it produces. In one regime — when both thresholds point outward — the carbon-minimizing network is faster *and* more dispersed at once; this is a sufficient condition, not a universal law.

The remainder of the paper proceeds as follows. Section 2 positions the paper relative to the location–inventory, carbon-network-design, and inventory–transport–emissions literatures. Section 3 develops the model, Section 4 establishes the results, Section 5 reports computational experiments, and Section 6 draws out the managerial implications.

2 Related Work

This study connects four streams: location–inventory design with risk pooling, carbon-aware network design, the joint control of inventory and transport mode under emissions regulation, and carbon as a priced operational lever in service-constrained inventory systems. Each develops an element of the two substitutions examined here — availability (inventory versus replenishment speed) and proximity (facilities versus last-mile travel). Each of these streams is well developed, and we draw on all four; we position the contribution by what it adds to each: the joint design of network density and replenishment speed under a carbon cap and a service requirement, together with the closed-form thresholds that relate the carbon-minimizing configuration to the cost-minimizing one.

2.1 Location–Inventory Design and Risk Pooling

A first stream establishes how demand risk pooling shapes distribution network design: jointly choosing facility location and safety stock, it characterizes the cost-optimal number and placement of distribution centers. Eppen (1979) characterizes the effect of centralization on expected cost in the multi-location newsvendor, establishing the square-root relationship between the number of stocking locations and pooled safety stock. Daskin et al. (2002) and Shen et al. (2003) embed working and safety stock into the facility-location decision, so that risk pooling shapes the optimal

number and placement of distribution centers, with extensions to stochastic settings by Snyder et al. (2007), to capacitated warehouses by Özsen et al. (2008), and to continuous-approximation allocation by Tsao et al. (2012). Daganzo (2005) provides the continuous-approximation foundation in which last-mile distance scales with the inverse square root of the number of facilities. Closest to our spatial–inventory coupling, Straubert (2024) ties the number of stocking locations to an exact reorder-point/order-quantity inventory model through a delivery-lead-time penalty, solved by continuous approximation. The centralize-versus-decentralize question these models pose can also turn on forces beyond demand pooling: Schmitt et al. (2015) show analytically that supply-disruption risk diversification can favor decentralization for a risk-averse firm, in contrast to the classical pooling result. These works develop network density and safety-stock placement under risk pooling, with a cost or service objective and no carbon; replenishment speed is taken as given, or, where lead time appears, fixed by exogenous fulfillment. The present study retains the risk-pooling structure and the density decision and adds an endogenous replenishment speed together with a carbon objective.

2.2 Carbon-Aware Network Design

A second stream brings carbon into facility location and network design, characterizing how an emissions objective or cap can change the optimal network configuration. Cachon (2014) relates retail store density to greenhouse-gas cost, with consumer travel to stores as the dominant emission channel and inventory entering through retail floor space. Turken et al. (2017) examine facility location and capacity under a carbon tax and emission limits, weighing centralization against decentralization through production scale economies; Harris et al. (2011) compare cost-optimal and emissions-aware network designs numerically; Wang et al. (2020) develop a carbon-compliant location–inventory model under demand and carbon-price uncertainty; Li et al. (2017) jointly optimize distribution-network design and two-echelon inventory under a CO_2 emission tax; and Hacardiaux and Tancrez (2020) assess the transport-emission savings of a shared, consolidated location–inventory network under horizontal cooperation. These studies make density or facility configuration the central decision and bring emissions into the objective, the constraints, or an ex-post assessment; transport enters through discrete modes or lead-time parameters within numerical mixed-integer programs, not as a continuously optimized replenishment speed. The present study treats network density and replenishment speed as joint decisions and resolves the proximity substitution between facilities and last-mile travel analytically.

2.3 Inventory, Transport Mode, and Emissions

A third stream couples transport-mode and lead-time choices with inventory control under emissions regulation, characterizing how carbon reshapes the trade-off between freight and the inventory it serves. Hoen et al. (2014) and Chen and Wang (2016) study transport mode selection alongside inventory under stochastic demand and emissions regulation; Konur and Schaefer (2014) solve the economic order quantity model with less-than-truckload and truckload carriers under

carbon cap, tax, and cap-and-trade regulation, and Schaefer and Konur (2015) extend this to a continuous-review, reorder-point inventory system under stochastic demand, characterizing the cost–emission Pareto frontier and analyzing lead-time and demand-variance sensitivity; and Drent et al. (2023) analyze dynamic dual supply-mode selection under an emission target; and Hammami et al. (2015) bring carbon from manufacturing, ordering, and inventory into a multi-echelon production–inventory model with lead-time constraints. Most of these mode-choice inventory models are single-location, while Hammami et al. (2015) is multi-echelon; in all of them lead time enters as a parameter, a constraint, or a discrete-mode outcome — in Schaefer and Konur (2015) it is a sensitivity and "green-investment" target rather than a decision — so replenishment speed is not a continuous decision chosen jointly with network density, and none derives a speed–density threshold. A discrete-mode predecessor of the present study, Boone and Ganeshan (2025), also sits in this stream: it chooses a discrete freight mode against inventory under an emission cap at a single location, with no network-density decision. The present study extends that mode choice to a continuous replenishment speed, prices carbon on the holding as well as the transport side, and embeds it in a network whose density is chosen jointly — yielding a continuous speed–density threshold and the complementarity that a single-location, discrete-mode model cannot express.

2.4 Carbon Policy, Service, and Safety-Stock Placement

A fourth stream prices carbon directly into inventory and service decisions, showing how a carbon price or cap shifts order quantities and safety-stock placement. Hua et al. (2011) and Bouchery et al. (2012) develop carbon-aware order-quantity models under cap-and-trade and multi-objective formulations, and Benjaafar et al. (2013) position cap, tax, and cap-and-trade as priced operational levers in simple supply-chain models; Chen et al. (2013) derive the closed-form carbon-constrained economic order quantity, the single-location analogue of the marginal carbon-intensity logic we extend to a network, and Arıkan and Jammernegg (2014) impose a product carbon-footprint constraint on a dual-sourcing newsvendor. Silbermayr et al. (2017) bring environmental constraints into inventory pooling with dependent demand. Within multi-echelon service models, Ni and Shu (2015) study the trade-off between service time and carbon in safety-stock placement under the guaranteed-service framework, and Wang et al. (2025) incorporate cap-and-trade into guaranteed-service placement. Probabilistic carbon limits have been handled through chance constraints in lot sizing (Koca, 2021) and in supply-chain network design (Shaw et al., 2016), typically inside large mixed-integer programs; we instead obtain a closed-form deterministic equivalent that scales the transport bucket by a single factor. A recent review (Sebatjane, 2025) surveys carbon-aware inventory models and the regulatory instruments they treat; across this body of work the literature remains fragmented across transport-emission modeling, service and inventory, network design, and demand or carbon uncertainty, and we bring these strands together. The service-placement models in particular price carbon against service and safety stock, with the joint design of network density and replenishment speed outside their scope. Prior work thus covers each of these ingredients — carbon-aware density, mode or speed choice, lead-time effects, service-constrained placement,

and chance-constrained carbon — but not their combination in a parsimonious analytical model in which network density and a continuous replenishment speed are joint decisions and the cost–carbon comparison is expressed through closed-form thresholds. The present study couples an endogenous replenishment speed with the network-density decision under a carbon cap and a service requirement and derives marginal carbon-intensity thresholds: a speed threshold comparing the carbon intensity per dollar of holding with that of inbound transport, a pooling-adjusted density threshold comparing last-mile intensity with a weighted average of the facility and holding intensities, and the complementarity that links them.

3 The Model

3.1 Operational environment

Region and markets. A firm serves N markets distributed over a service region of area A. Per-period demand at market i is a random variable with mean μ and standard deviation σ, identically distributed across markets with common pairwise correlation $\rho \in [0, 1)$. Demand is stationary and the product is single. Markets and distribution centers are treated symmetrically, the standard simplification that admits closed-form continuous-approximation results (Daganzo, 2005).

Distribution centers and geography. The firm opens n identical distribution centers (DCs), $1 \leq n \leq N$, each serving a sub-region of area A/n. By the continuous-approximation spatial law, the mean last-mile (DC-to-market) distance falls as the square root of the served area,

$$d(n) \;=\; \delta\, n^{-1/2}, \qquad \delta \propto \sqrt{A}, \tag{1}$$

so a denser network shortens delivery but only at a decreasing rate (Daganzo, 2005).

Replenishment and the speed lever. Each DC is replenished directly from the source at a chosen speed $v \in [v_{\min}, v_{\max}]$. Individual DCs sit at different distances from the source; what enters cost and carbon is total inbound freight, $\sum_j (\text{throughput}_j)\, D_j = N\mu\, \bar{D}$, the total throughput times the *mean* source-to-DC distance $\bar{D}$, since the heterogeneous distances enter only through this average once throughput is conserved at $N\mu$. Because the DCs fill a region of fixed size served from a fixed source, $\bar{D}$ is set by the region's geometry (order $\sqrt{A}$) and is approximately invariant to n; we write $D \equiv \bar{D}$ and treat it as constant.[1] The replenishment lead time is then

$$L(v) \;=\; D/v. \tag{2}$$

[1] Two modeling reductions are embedded here. First, the symmetric-DC reduction assigns every DC the representative lead time $L = D/v$; since safety stock is concave in L, using $\bar{D}$ in the $\sqrt{L}$ term of (4) is a mild Jensen approximation of the heterogeneous distances. Second, replenishment is point-to-point; were the DCs served on a consolidated route, inbound distance would grow as $\sqrt{n}$ in the manner of Cachon (2014)'s store-delivery tour, adding a term that reinforces centralization in Proposition 4 but leaving the speed result of Proposition 2 unchanged.

Faster replenishment is simultaneously more expensive and more carbon-intensive. The maintained assumption is only that the inbound cost rate $t(v)$ and emission rate $\varepsilon(v)$ (both per unit per mile) are increasing and convex in v; for the closed-form results we use the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$ and $t(v) = t_0 + t_1 v^2$, the canonical form motivated by the cubic fuel-speed law of slow steaming (Psaraftis and Kontovas, 2013; Cariou, 2011). We read v as a general index of replenishment speed — vessel speed, or equivalently a faster and more carbon-intensive mode — so slow steaming is a motivating special case, not the only interpretation. Treating replenishment speed as a continuously optimized decision, chosen jointly with network density — the element absent from carbon network-design models, which represent transport through discrete modes or lead-time parameters — is what couples the transport and holding buckets in this distribution setting.

Inventory and service. Each DC operates a base-stock policy and must meet a cycle service level α — the in-stock probability at the end of a period, the Type-1 (cycle) service measure. With normally distributed demand the safety factor is $z = \Phi^{-1}(\alpha)$, taken here as a service requirement (a parameter). A DC that pools $g = N/n$ markets faces aggregate per-period demand with standard deviation $\sigma\sqrt{g\,[\,1 + (g-1)\rho\,]}$, the statistical economy of scale in inventory (Eppen, 1979). Covering lead-time demand, each DC holds safety stock $z\,\sigma\sqrt{L}\,\sqrt{g[1 + (g-1)\rho]}$. Summing over the n DCs and writing the *pooling expression*

$$\Psi(n) \;=\; N\rho + n(1-\rho), \tag{3}$$

the system-wide safety stock is

$$\mathrm{SS}(n, v) \;=\; z\,\sigma\,\sqrt{D/v}\;\sqrt{N\,\Psi(n)}. \tag{4}$$

At $\rho = 0$, (4) reduces to $z\sigma\sqrt{L}\sqrt{nN}$, the square-root law: decentralizing from 1 to N DCs multiplies safety stock by $\sqrt{N}$. At $\rho = 1$, $\Psi(n) = N$ and safety stock is invariant to n — pooling buys nothing.

Decisions. The firm chooses the network density $n \in [1, N]$ (relaxed to a continuous variable and rounded) and the replenishment speed $v \in [v_{\min}, v_{\max}]$. Service is a requirement, so α (hence z) is a parameter in the core model. Table 1 collects the notation.

3.2 Cost and carbon

Before stating the cost and carbon functions we fix their scope. Both are flows defined over a single representative period, consistent with the stationary single-product environment of Section 3: F, φ are charged per DC per period, h, η per unit held per period, and the transport terms accrue against the per-period throughput. Throughput itself is conserved at $N\mu$ regardless of the configuration: the firm moves the mean demand of all N markets through the network whatever density n and speed v it chooses. For this reason we set aside the cost and carbon of production (and of the

Table 1: Notation. The decision variables are the network density n and the replenishment speed v; all other quantities are exogenous parameters or derived.

Symbol	**Meaning**
Decision variables	
n	number of DCs (network density), $1 \le n \le N$
v	replenishment speed, $v \in [v_{\min}, v_{\max}]$
Demand (exogenous)	
N	number of markets
$\mu,\ \sigma$	mean, standard deviation of demand per market per period
ρ	cross-market demand correlation, $\rho \in [0, 1)$
Geography, cost, carbon (exogenous)	
$A,\ D,\ \delta$	region area; inbound distance; last-mile distance constant
$F,\ \varphi$	fixed cost, carbon per DC per period
$h,\ \eta$	holding cost, carbon per unit per period
$t(v),\ \varepsilon(v)$	inbound transport cost, carbon per unit-mile (convex in v)
$t_L,\ \varepsilon_L$	last-mile transport cost, carbon per unit-mile
α	cycle service level (in-stock prob.); $z = \Phi^{-1}(\alpha)$
$\tau,\ B$	carbon price; carbon cap
Derived	
$g = N/n$	markets served per DC
$L = D/v$	replenishment lead time
$d(n) = \delta\, n^{-1/2}$	mean last-mile distance
$\Psi(n) = N\rho + n(1-\rho)$	pooling expression
$\mathrm{SS}(n, v)$	system-wide safety stock, eq. (4)

product's embodied content): these scale with $N\mu$ and are therefore invariant to the two decisions, so including them would add a constant to (5)–(6) that shifts levels without affecting either the location of the optimum or the cost–carbon trade-off. We retain the four pieces the decisions do move — a fixed facility term, a holding term, an inbound transport term, and a last-mile transport term — and model cost and carbon as the same four pieces, with dollar rates in one and kilogram rates in the other. With per-period throughput $N\mu$,

$$K(n, v) = \underbrace{F\, n}_{\text{fixed}} + \underbrace{h\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{t(v)\, N\mu D}_{\text{inbound}} + \underbrace{t_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{5}$$

$$C(n, v) = \underbrace{\varphi\, n}_{\text{fixed}} + \underbrace{\eta\, \mathrm{SS}(n, v)}_{\text{holding}} + \underbrace{\varepsilon(v)\, N\mu D}_{\text{inbound}} + \underbrace{\varepsilon_L\, N\mu\, d(n)}_{\text{last-mile}}, \tag{6}$$

where F, φ are fixed cost/carbon per DC, h, η are holding cost/carbon per unit per period, and t_L, ε_L are last-mile cost/carbon rates (Shen et al., 2003; Hua et al., 2011; Benjaafar et al., 2013). Each term is computed by multiplying a rate by the physical quantity it prices, and the two functions differ only in whether that rate is in dollars or in kilograms. We take the four terms in turn, in the order facilities, holding, inbound, last-mile, and then collect them into the two buckets used in the analysis.

Facilities. The first term is the periodic charge of operating the network. On the cost side $F\,n$ scales the per-DC fixed cost F — the amortized lease or capital recovery, baseline staffing, and the building's fixed operating expense — by the number of open DCs; on the carbon side $\varphi\,n$ scales the per-DC facility carbon φ the same way. With F in dollars per DC per period and φ in kilograms per DC per period, the cost term carries units of dollars per period and the carbon term kilograms per period. Both rise linearly in n and do not depend on v: each additional DC adds a full increment F (resp. φ). The facility-side carbon φ and the holding-side carbon η below are distinct and not double-counted: φ is the baseline energy of operating a DC (building lighting and climate control, amortized embodied carbon), independent of the inventory held, whereas η in the holding term is the additional energy per unit stored (refrigeration, handling).

Holding. The second term values the safety stock carried across the network. It multiplies the system-wide safety stock $\mathrm{SS}(n, v)$ of (4) by the holding cost rate h (dollars per unit per period) in K and by the holding carbon rate η (kilograms per unit per period) in C, so the term is in dollars per period and kilograms per period respectively. By (4) the stock itself is $\mathrm{SS}(n, v) = z\,\sigma\,\sqrt{D/v}\,\sqrt{N\,\Psi(n)}$, so the term is computed from the service factor z, the per-market demand variability σ, the lead time $L = D/v$ of (2) induced by the source-to-DC distance D and the chosen speed v, and the pooling expression $\Psi(n) = N\rho + n(1-\rho)$ of (3). The term rises in n and falls in v: through $\Psi(n)$ it is increasing and concave in n, so spreading inventory over more DCs forgoes the statistical economy of scale (Eppen, 1979), the more so the lower the correlation ρ; and through the $\sqrt{D/v}$ factor it falls in v as $\sqrt{1/v}$, since faster replenishment shortens the lead time the safety stock must cover. This is the channel through which the speed lever enters the holding side and couples it to inbound transport.

Inbound transport. The third term is the cost and carbon of moving goods from the source to the DCs. It multiplies the per-unit-mile inbound rate — $t(v)$ in dollars per unit per mile for cost, $\varepsilon(v)$ in kilograms per unit per mile for carbon — by the inbound freight task $N\mu D$, the per-period throughput $N\mu$ carried over the representative source-to-DC distance D, so the term is again dollars (resp. kilograms) per period. Since D is set by the region's geometry and treated as approximately invariant to n, the freight quantity $N\mu D$ does not move with network density and the term varies through the rate. Both $t(v)$ and $\varepsilon(v)$ are increasing and convex in v — in the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$ and $t(v) = t_0 + t_1 v^2$, the canonical form motivated by the cubic fuel-speed law of slow steaming (Psaraftis and Kontovas, 2013; Cariou, 2011) — so the inbound term rises, at an increasing rate, in v and opposes the holding term, which falls in v.

Last-mile transport. The fourth term is the cost and carbon of the final DC-to-market leg. It multiplies the per-unit-mile last-mile rate — t_L in dollars per unit per mile for cost, ε_L in kilograms per unit per mile for carbon — by the delivery task $N\mu\,d(n)$, the same throughput $N\mu$ carried over the mean last-mile distance $d(n) = \delta\,n^{-1/2}$ of (1), so the term is again dollars (resp. kilograms) per period. Unlike inbound freight, this term depends on n through $d(n)$: a denser network shortens the

average delivery distance as the square root of the served area, so the last-mile term is decreasing and convex in n (Daganzo, 2005). A constant per-mile last-mile rate leaves it independent of v, which acts only on the inbound leg. The last-mile term therefore decreases in n, partially offsetting the increase in the facility term.

Decomposition and scope. The carbon in (6) splits into a *transport bucket* ($\varepsilon(v)N\mu D + \varepsilon_L N\mu\, d(n)$, rising in v, falling in n) and a *holding bucket* ($\varphi n + \eta\,\mathrm{SS}$, falling in v, rising in n); the same partition organizes the cost in (5), with the inbound and last-mile terms forming the transport bucket and the fixed and holding terms the holding bucket. The tension that organizes the analysis is that both ways to shrink the transport bucket — slow down ($v \downarrow$) or spread out ($n \uparrow$) — inflate the holding bucket; treating replenishment speed as a decision, which couples the inbound transport term to the holding term through the lead time $L = D/v$, is what links the two buckets in this distribution setting. Two further points about scope frame what the four terms cover. The inventory side carries safety stock through $\mathrm{SS}(n, v)$. Cycle (working) stock, about $\mu/2$ per DC per review period under a base-stock policy, is invariant to both n and v and enters as a constant; in-transit (pipeline) inventory, $N\mu L = N\mu D/v$, does vary with v and is omitted from the core, where we show in Section 4 that including it at the warehouse holding rates leaves the speed-reversal threshold unchanged, though the level of v^* shifts. And, as noted above, production and procurement cost is held outside (5)–(6): with throughput conserved at $N\mu$, the quantity sourced and produced is invariant to n and v, so it enters as a constant that does not affect either lever and is set aside for the optimization.

3.3 The program

The firm minimizes cost subject to a carbon cap B and the service requirement,

$$\min_{1\le n\le N,\ v_{\min}\le v\le v_{\max}} K(n, v) \quad \text{s.t.} \quad C(n, v) \le B. \tag{7}$$

Here K and C are expected per-period flows: under the stationary demand of Section 3 they are the long-run average cost and carbon of a configuration, so the cap in (7) bounds *expected* carbon, the form annual carbon accounting and most regulation take. Because demand is random, realized carbon fluctuates about $C(n, v)$; we develop the analysis for the expected-value cap and return below to a chance-constrained cap that bounds realized carbon with prescribed confidence (Proposition 7).

Equivalently, pricing carbon at $\tau \ge 0$ gives the *priced objective* $J = K + \tau C$ — total cost plus priced carbon — which the firm minimizes over n and v; by Lagrangian duality the cap multiplier equals the price ($\lambda = \tau$) at an interior optimum, as in the carbon-constrained inventory models of Chen et al. (2013), and sweeping τ from 0 to ∞ traces the cost–carbon Pareto frontier. Because J can be non-convex in n (the pooling term is concave) and n is ultimately integer, the price–cap correspondence $\lambda = \tau$ holds within a regime where the optimal configuration is interior and stable;

across a regime switch the frontier has a kink and the correspondence can jump. Because (5)–(6) share the four-piece structure, the priced objective is the *same* four pieces with each rate blended, $\widetilde{\text{rate}} = \text{rate}_\$ + \tau\,\text{rate}_{\text{kg}}$:

$$J(n, v; \tau) \;=\; a\,n \;+\; b(v)\,\sqrt{\Psi(n)} \;+\; c(v) \;+\; e\,n^{-1/2}, \tag{8}$$

with $a = F+\tau\varphi,\ b(v) = (h+\tau\eta)\,z\sigma\sqrt{N}\,\sqrt{D/v},\ c(v) = (t(v)+\tau\varepsilon(v))\,N\mu D$, and $e = (t_L+\tau\varepsilon_L)\,N\mu\delta$. Setting $\tau = 0$ recovers cost; $\tau \to \infty$ recovers carbon.

4 Analysis

We now analyze the firm's problem to characterize the decisions it implies and the logic a manager would use to make them. Two structural facts carry the argument: cost and carbon are the same four pieces, so a carbon price acts through a small set of ratios; and the two levers — the firm's two substitutions, inventory versus speed for availability and facilities versus last-mile travel for proximity — are coupled, so a carbon-minimizing chain is redesigned on both at once rather than tuned on one.

When a carbon constraint binds, the firm must decide how fast to replenish, how to size its network, whether to move one lever or both, and which way a given product's configuration should shift. The results below take these decisions in turn: holding the network fixed, Proposition 2 characterizes the replenishment speed and Proposition 3 ties it to the service level and demand variability; holding speed fixed, Proposition 4 characterizes the density; Proposition 5 shows the two levers are coupled; and Corollary 6 combines the comparisons, whose direction is set throughout by the carbon-intensity-per-dollar ratios of Lemma 1.

Lemma 1 (Carbon intensity per dollar)**.** *Pricing carbon at τ replaces each cost rate in* (5) *by the blended rate* $\text{rate}_\$ + \tau\,\text{rate}_{\text{kg}}$*, yielding the objective J of* (8)*. In the interior fixed-lever comparisons below, the* direction *in which the carbon-minimizing choice departs from the cost-minimizing one is governed by four carbon-intensity-per-dollar ratios — holding η/h, inbound transport ε_0/t_1, last-mile ε_L/t_L, and facilities φ/F — while the magnitude and the realized joint optimum also depend on pooling, the bounds, and the complementarity of the two levers.*

Proof. Collecting (5)–(6), the coefficient of each of the four terms of $K + \tau C$ is $\text{rate}_\$ + \tau\,\text{rate}_{\text{kg}}$, giving (8) with a, b, c, e as stated. For the speed lever the cost optimum ($\tau = 0$) and carbon optimum ($\tau \to \infty$) coincide iff the two rate pairs entering it — holding and inbound transport — are proportional, i.e. iff $\eta/h = \varepsilon_0/t_1$ (Proposition 2). The density lever combines three terms — facilities, holding, and last mile — so its comparison is governed not by a single two-pair proportionality but by the pooling-adjusted threshold of Proposition 4, a weighted average of φ/F and η/h. For the joint problem, proportionality of all four ratios is sufficient for the two optima to coincide but is not shown to be necessary. □

The carbon-minimizing choice on each lever therefore depends on each activity's carbon intensity *relative to its cost*, not on its absolute rates.

Differentiating J in (8), with $\Psi'(n) = 1 - \rho$ and $b'(v) = -b(v)/2v$, gives the stationarity conditions

$$a + \frac{b(v)\,(1-\rho)}{2\sqrt{\Psi(n)}} = \frac{e}{2}\,n^{-3/2}, \tag{9}$$

$$\big(t'(v) + \tau\varepsilon'(v)\big) N\mu D = \frac{b(v)\,\sqrt{\Psi(n)}}{2v}, \tag{10}$$

the first balancing the marginal cost of opening a DC — the facility term a plus the lost-pooling holding term, scaled by $(1-\rho)$ so it vanishes when pooling is useless — against the marginal last-mile saving; the second balancing the marginal transport cost-and-carbon of speed against the marginal holding saving it buys.

Availability: inventory or speed. Hold the network at any density n and ask how fast to replenish; the speed subproblem is convex.

Under a carbon cap or guidance to adopt slower modes, the transport-focused view is that slowing replenishment lowers the transport component of distribution carbon. Whether it lowers the total is less clear, because the longer lead time enlarges the safety stock, whose holding is itself carbon-intensive, so one source rises while the other falls.

Proposition 2 (Speed reversal)**.** *Fix the density n and write the speed-dependent cost and carbon as*

$$K_v(v) = h\,A(n)\,v^{-1/2} + T\,t(v), \qquad C_v(v) = \eta\,A(n)\,v^{-1/2} + T\,\varepsilon(v),$$

with $A(n) = z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}$ and $T = N\mu D$. If t and ε are increasing, differentiable, and convex and each objective is strictly convex in v, then at a common density and for interior optima the carbon-minimizing speed exceeds the cost-minimizing speed if and only if holding is more carbon-intensive per dollar than inbound transport at the margin*,*

$$\frac{\eta}{h} \;>\; \frac{\varepsilon'(v_{\mathrm{K}})}{t'(v_{\mathrm{K}})}, \tag{11}$$

evaluated at the cost-minimizing speed v_{K}. With the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$, $t(v) = t_0 + t_1 v^2$, the threshold reduces to $\eta/h > \varepsilon_0/t_1$ and the unconstrained optimum is closed form,

$$v^*(n) \;=\; \left[\frac{(h+\tau\eta)\,z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}}{4\,(t_1+\tau\varepsilon_0)\,N\mu D}\right]^{2/5}. \tag{12}$$

On $[v_{\min}, v_{\max}]$ the optima are the projections of the unconstrained speeds onto the interval; under the threshold (11) *the comparison $v_C \geq v_K$ then holds weakly, and strictly when neither projection binds.*

Proof. $T\,t(v)$ and $T\,\varepsilon(v)$ are convex and $A(n)v^{-1/2}$ has $\partial_v^2 = \frac{3}{4}A(n)v^{-5/2} > 0$, so both objectives are strictly convex in v. The cost optimum satisfies $K_v'(v_{\mathrm{K}}) = 0$, i.e. $T\,t'(v_{\mathrm{K}}) = \frac{1}{2}h\,A(n)\,v_{\mathrm{K}}^{-3/2}$, so that $\frac{1}{2}A(n)v_{\mathrm{K}}^{-3/2} = T\,t'(v_{\mathrm{K}})/h$. Evaluating the carbon derivative there,

$$C_v'(v_{\mathrm{K}}) = -\eta\,\tfrac{1}{2}A(n)v_{\mathrm{K}}^{-3/2} + T\,\varepsilon'(v_{\mathrm{K}}) = T\big(\varepsilon'(v_{\mathrm{K}}) - (\eta/h)\,t'(v_{\mathrm{K}})\big),$$

which is negative — so $v_C > v_{\mathrm{K}}$ by strict convexity — if and only if (11). With $t' = 2t_1 v$ and $\varepsilon' = 2\varepsilon_0 v$ the marginal ratio is ε_0/t_1, and the priced first-order condition (10) gives $v^{5/2} = (h{+}\tau\eta)z\sigma\sqrt{ND}\,\sqrt{\Psi}/[4(t_1{+}\tau\varepsilon_0)N\mu D]$, which is (12). The projection and the boundary qualification are immediate. □

The economics is a U-shape: faster replenishment is more carbon-intensive in transport but less inventory-intensive, because it shrinks the safety stock that a longer lead time would require. For a product whose inventory is the emissions-intensive component — refrigerated, volatile, or held to a high service level — the carbon-minimizing response to a tighter cap is to ship *faster*. This is a distinct configuration from the one the slow-steaming literature describes (Psaraftis and Kontovas, 2013; Cariou, 2011), and it extends the finding that shorter lead times lower both inventory cost and emissions (Schaefer and Konur, 2015) by pricing the transport carbon that speed costs, which turns a one-directional saving into an interior optimum. The comparison is stated at a common density; at the joint optimum the densities differ, but since v^* rises in n (Proposition 5) the density channel reinforces the reversal when carbon also disperses the network and only partly offsets it otherwise.

Setting a cycle-service target, or accepting a more volatile item, is ordinarily treated as an inventory-cost decision; the same closed form shows that it also moves the carbon-minimizing configuration and so belongs in the carbon design rather than only the cost budget.

Proposition 3 (Service and volatility are carbon parameters)**.** *$v^*(n)$ is increasing in the safety factor $z = \Phi^{-1}(\alpha)$ and in the demand standard deviation σ, and both raise the holding term in* (9)*. At an interior reduced optimum — with $J^{\mathrm{r}}(n;s) \equiv J\big(n, v^*(n;s); s\big)$ the objective along the speed response and reduced second-order condition $J^{\mathrm{r}}_{nn} > 0$ — this is a local comparative-static result: a higher z or σ raises the inventory penalty of a denser network and moves the carbon-minimizing density toward consolidation.*

Proof. From (12), $v^* \propto (z\sigma)^{2/5}$, so z and σ raise the optimal speed. Because that speed increase by itself lowers the holding term, the net effect on density must be read along the speed response. Because the speed condition (10) pins down $v^*(n;s)$ for each n independently of the density condition, we may substitute it into the density condition before differentiating; the resulting reduced first-order condition $J^{\mathrm{r}}_n(n;s) = J_n\big(n, v^*(n;s); s\big) = 0$ needs no separate envelope term in n (since (10), i.e. $J_v = 0$, holds identically along $v^*(n;s)$), and its total s-derivative already carries the induced speed change inside $v^*(n;s)$. In this reduced condition (with $s = z\sigma$) the holding term is $b(v^*(n))(1-\rho)/(2\sqrt{\Psi(n)})$ with $b(v^*(n)) \propto s\,(v^*)^{-1/2} \propto s^{4/5}$, increasing in s even after the

induced rise in speed. By the implicit function theorem applied to this reduced condition, and with the reduced second-order condition $J^{\mathrm{r}}_{nn} > 0$ — the curvature in n of the reduced objective $J^{\mathrm{r}}(n;s) = J\big(n, v^*(n;s); s\big)$, assumed to hold locally and verified at the calibrated instances — one has $\mathrm{d}n^*/\mathrm{d}s = -J^{\mathrm{r}}_{ns}/J^{\mathrm{r}}_{nn} < 0$, since the only s-dependent term, $\propto s^{4/5}$, raises J^{r}_n. Thus a higher service factor or volatility raises the carbon-minimizing speed and lowers the carbon-minimizing density; the consolidation force survives the induced speed increase rather than being offset by it. □

A service promise is therefore not only an inventory-cost decision. Raising a product's cycle-service target from 95% to 99%, or carrying a more erratic item, enlarges safety stock and so — under carbon — raises its optimal replenishment speed and, in the network decision, strengthens the pull toward consolidation (more pooling) — a distinct force from the dispersing force of demand correlation. Service level and demand variability belong in the carbon design, not only the cost budget; treating the service target as a third decision, set jointly with speed and density, is a natural extension (Section 7).

Proximity: facilities or last-mile travel. Now hold speed fixed and choose the density; pooling makes this problem less uniform than the speed problem.

Holding speed fixed, the firm chooses how many DCs to operate, and consolidation carries two carbon effects with different signs: it pools demand and saves facility energy, but it lengthens the last mile.

Proposition 4 (Density and dispersion)**.** *Fix the speed v and let*

$$G(n,v) \;=\; \frac{\partial\,\mathrm{SS}(n,v)}{\partial n} \;=\; z\sigma\sqrt{D/v}\,\sqrt{N}\,\frac{1-\rho}{2\sqrt{\Psi(n)}} \;\geq\; 0 \tag{13}$$

be the marginal safety-stock penalty of an additional DC. The priced objective $J(\cdot, v)$ is strictly convex in n when

$$3e\,n^{-5/2} \;>\; b(v)(1-\rho)^2\,\Psi(n)^{-3/2} \qquad \textit{for all } n \in [1,N], \tag{14}$$

in which case the stationary point is unique; the minimizer is interior if and only if, in addition, $J_n(1) < 0 < J_n(N)$, and otherwise lies at a boundary, $n^* \in \{1, N\}$. Under this condition, and for interior cost and carbon density optima, the carbon-minimizing density exceeds the cost-minimizing density, $n_C(v) > n_K(v)$, if and only if last-mile transport is more carbon-intensive per dollar than the pooling-adjusted facility threshold*,*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\,G\big(n_K(v), v\big)}{F + h\,G\big(n_K(v), v\big)}. \tag{15}$$

The right-hand side is a weighted average of the facility ratio φ/F and the holding ratio η/h, with weight on η/h rising in the pooling penalty G: it equals φ/F when pooling is absent and

tilts monotonically toward η/h as G grows (lower correlation, higher service factor or volatility, or slower replenishment). In the no-pooling limit ($\rho = 1$, $G = 0$) the optimum is closed form, $n^* = (e/2a)^{2/3}$, and (15) reduces to

$$\frac{\varepsilon_L}{t_L} > \frac{\varphi}{F}. \tag{16}$$

The ordering $n_C(v) > n_K(v)$ is strict at interior optima; when an optimum lies at a boundary ($n = 1$ or N) it holds as a projected weak ordering.

Proof. With $\frac{d^2}{dn^2}\sqrt{\Psi} = -\frac{(1-\rho)^2}{4}\Psi^{-3/2}$, $\partial_n^2 J = \frac{3}{4}e\,n^{-5/2} - \frac{1}{4}b(v)(1-\rho)^2\Psi^{-3/2}$, positive on $[1, N]$ exactly under (14), so $J(\cdot, v)$ is then strictly convex with a unique stationary point, interior iff $J_n(1) < 0 < J_n(N)$. For the threshold, take $\tau = 0$ for cost and $\tau \to \infty$ for carbon. The cost density satisfies $\partial_n K = F + h\,G(n_K, v) - t_L N\mu\delta/(2n_K^{3/2}) = 0$, so the last-mile marginal $t_L N\mu\delta/(2n_K^{3/2}) = F + h\,G(n_K, v)$. Evaluating the carbon derivative there,

$$\partial_n C\big|_{n_K} = \varphi + \eta\,G(n_K, v) - \frac{\varepsilon_L N\mu\delta}{2n_K^{3/2}} = \varphi + \eta\,G(n_K, v) - \frac{\varepsilon_L}{t_L}\left[F + h\,G(n_K, v)\right],$$

which is negative — so $n_C > n_K$ by convexity — if and only if (15). Writing $R(G) = (\varphi + \eta G)/(F + hG)$, we have $R(0) = \varphi/F$, $R(G) \to \eta/h$ as $G \to \infty$, and $R'(G) = (\eta F - h\varphi)/(F + hG)^2$, so R is monotone with $\operatorname{sign} R' = \operatorname{sign}(\eta/h - \varphi/F)$ and lies between φ/F and η/h. At $\rho = 1$, $G = 0$ and $\Psi \equiv N$, so (9) gives $n^* = (e/2a)^{2/3}$ and (15) becomes (16). □

The density decision is the centralize-versus-decentralize question in carbon terms. Centralizing pools demand, so safety stock falls with the square-root law (Eppen, 1979), and it consolidates facility energy, while *lengthening* the last mile. By (15) the carbon-minimizing network is more dispersed than the cost-minimizing one exactly when the last mile's carbon intensity per dollar, ε_L/t_L, exceeds a facility threshold that pooling shifts from φ/F toward the holding ratio η/h. Pooling is thus not a vague inward force: it enters the threshold as a weight G on η/h, so strong pooling (low correlation, high service or volatility) raises the bar for dispersion when holding is itself carbon-intensive, and lowers it otherwise. The classical location–inventory trade-off weighs the pooling economy against proximity (Daskin et al., 2002; Shen et al., 2003); pricing carbon reweights it, and for a bulky, low-value good carried by diesel van the last-mile ratio clears the threshold, so the carbon-minimizing network holds more inventory in facilities closer to demand — the firm-operated counterpart of Cachon (2014), there through consumer travel and here through firm last-mile delivery. A corner optimum, when it arises, is itself well defined: fully centralize, or fully decentralize.

Complementarity of speed and density. The speed and density choices do not separate.

A firm pursuing abatement might run a single program — a slow-steaming mandate that sets speed, or a consolidation initiative that sets density — which raises the question whether either lever can be tuned in isolation or whether the carbon-minimizing speed and density must be chosen together.

Proposition 5 (Complementarity)**.** *$v^*(n)$ is increasing in n, and under the density second-order condition $n^*(v)$ is increasing in v.*

Proof. From (12), $v^* \propto \Psi(n)^{1/5}$ with $\Psi'(n) = 1 - \rho \geq 0$, so $\partial v^*/\partial n \geq 0$, strict for $\rho < 1$. For the density response, apply the implicit function theorem to the first-order condition $J_n(n, v) = 0$ of (9): under the density second-order condition $J_{nn} > 0$ of (14), $\mathrm{d}n^*/\mathrm{d}v = -J_{nv}/J_{nn}$. With $b(v) = B_0 v^{-1/2}$ and $B_0 = (h + \tau\eta)z\sigma\sqrt{ND}$, the cross-partial $J_{nv} = b'(v)(1-\rho)/(2\sqrt{\Psi(n)}) = -b(v)(1-\rho)/(4v\sqrt{\Psi(n)})$ is negative for $\rho < 1$, so $\mathrm{d}n^*/\mathrm{d}v > 0$. □

Faster replenishment shrinks safety stock, lowering the inventory penalty of spreading out; a denser network raises the inventory at stake, raising the value of speed, so the two levers reinforce. The optimal degree of centralization is therefore contingent on replenishment speed: since $n^*(v)$ increases in v, a firm that can replenish faster can operate a more decentralized network at the same inventory burden, so an investment in speed and the choice of network density are best evaluated together. A program that adjusts only one — speed or density — attains part of the available abatement, whereas the carbon-minimizing response adjusts both.

The joint regime. Combining the two comparisons identifies when carbon points away from slow-and-consolidated on both axes at once.

For a single product under a tighter cap, the firm must decide which way to move each lever — faster or slower, more dispersed or more consolidated — and the two directions need not align. The carbon-intensity-per-dollar comparisons of Propositions 2 and 4 — holding against inbound transport, η/h versus ε_0/t_1, and last mile against facilities, ε_L/t_L versus φ/F — govern the direction on each axis; on density, when $\rho < 1$ the last-mile comparison is the outward force and the holding ratio η/h the inward one. The corollary below records the case in which both axes point outward.

Corollary 6 (Faster and more dispersed)**.** *Suppose, over the relevant range of configurations, the speed threshold of Proposition 2 holds at every density and the pooling-adjusted density threshold* (15) *of Proposition 4 holds at every speed, so that the carbon-minimizing best responses dominate the cost-minimizing ones pointwise, $v_C(n) \geq v_K(n)$ for all n and $n_C(v) \geq n_K(v)$ for all v. Then best-response iteration from the cost optimum converges to a carbon best-response equilibrium that is weakly faster and weakly more dispersed than the cost-minimizing configuration, strictly so at interior optima. At the calibrated instances of Section 5 the reported finite-grid search finds no configuration with a lower priced objective than this equilibrium.*

Proof. The two thresholds give the pointwise dominance $v_C(n) \geq v_K(n)$ and $n_C(v) \geq n_K(v)$ (Propositions 2 and 4), and by Proposition 5 each best response is nondecreasing ($v^*(\cdot)$ in n, $n^*(\cdot)$ in v). Starting from the cost optimum (n_K, v_K), a fixed point of the cost best responses, and iterating the carbon best responses gives a nondecreasing sequence: $v_C(n_K) \geq v_K(n_K) = v_K$, then $n_C\big(v_C(n_K)\big) \geq n_C(v_K) \geq n_K(v_K) = n_K$, and so on. The iterates are monotone and bounded, hence converge to a fixed point of the carbon best-response map that dominates (n_K, v_K) (Topkis, 1998), with strict inequalities at interior optima. This best-response dominance is the

content of the corollary and holds without further assumptions. Global optimality is not asserted in general: the priced objective need not be convex over the whole feasible box $[1, N] \times [v_{\min}, v_{\max}]$, because the speed–pooling cross term $J_{nv} = -b(v)(1-\rho)/(4v\sqrt{\Psi(n)})$ contributes a Hessian deficit $-\frac{1}{4}b(v)^2(1-\rho)^2 v^{-2}\Psi^{-1}$ that the own curvatures fail to cover in the extreme high-density, low-speed corner. □

At each calibrated instance of Section 5 we corroborate this equilibrium with a finite-grid search. On a rectangle R containing the cost optimum and the carbon equilibrium, the search finds no configuration with a lower priced objective: over the grid the priced Hessian is positive definite ($J_{nn} > 0$ and $J_{nn}J_{vv} - J_{nv}^2 > 0$), and the candidate beats every sampled boundary configuration, with $J_n(1) < 0 < J_n(N)$. This is a finite-grid verification for the calibrated instances, not a proof of global optimality over the continuous region; establishing the latter would require interval or analytic bounds on the Hessian determinant and the boundary values across R.

A refrigerated, volatile, high-service product delivered by diesel van illustrates this regime: its carbon-minimizing chain replenishes faster and holds stock closer to demand, rather than slower and more consolidated. We state this as a sufficient condition rather than a law — the contribution is not that this corner is universal but that the carbon-minimizing *direction* on each axis is set by a carbon-intensity-per-dollar comparison, and one of the four resulting regimes is this inverted one. Those two comparisons — holding against inbound transport, and last mile against facilities — sort any product into one of four regimes and indicate the direction in which carbon shifts its speed and its network. We draw out the resulting carbon strategy in Section 6.

Demand uncertainty: a chance-constrained cap. The cost and carbon functions are expected per-period flows, so the cap in (7) bounds *expected* carbon. Because demand is random, realized carbon fluctuates about $C(n, v)$: the facility charge and the base-stock safety-stock target are planned and treated as deterministic, but inbound and last-mile freight scale with realized throughput $Q = \sum_i X_i$, the sum of the realized market demands, which has mean $N\mu$ and variance $\sigma_Q^2 = \sigma^2 N\,[\,1 + (N-1)\rho\,]$. Realized carbon is then

$$\tilde{C}(n, v) \;=\; \varphi n + \eta\,\mathrm{SS}(n, v) + \big[\varepsilon(v)D + \varepsilon_L d(n)\big]\,Q, \tag{17}$$

and a firm that requires realized carbon to stay within B with confidence $1 - \beta_c$ — where β_c is the carbon-side tail probability, written with the subscript c to distinguish it from the inventory fill-rate β of the lot-sizing literature — imposes the chance constraint $\Pr\{\tilde{C}(n, v) \le B\} \ge 1 - \beta_c$, the carbon-side counterpart of the cycle service level α on the inventory side.

Proposition 7 (Deterministic equivalent of the chance constraint)**.** *This chance constraint is equivalent to*

$$\varphi n + \eta\,\mathrm{SS}(n, v) + m\big[\varepsilon(v)N\mu D + \varepsilon_L N\mu\, d(n)\big] \;\le\; B, \tag{18}$$

the carbon function (6) *with the transport bucket scaled by* $m = q_{1-\beta_c}/(N\mu)$*, where* $q_{1-\beta_c}$ *is the* $(1 - \beta_c)$ *quantile of realized throughput* Q*, and the facility and holding terms unchanged. Under*

the normal approximation $Q \sim \mathcal{N}(N\mu, \sigma_Q^2)$ *this is* $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$*, with* $z_{\beta_c} = \Phi^{-1}(1-\beta_c)$ *and* $\mathrm{CV}_Q = \sigma_Q/(N\mu)$*; then* $m \geq 1$ *whenever* $\beta_c \leq \frac{1}{2}$*.*

Proof. In (17) the throughput coefficient $g(n,v) = \varepsilon(v)D + \varepsilon_L d(n) > 0$, so $\{\tilde{C} \leq B\} = \{Q \leq (B - \varphi n - \eta \mathrm{SS})/g\}$, and $\Pr\{\tilde{C} \leq B\} \geq 1 - \beta_c$ holds iff $(B - \varphi n - \eta \mathrm{SS})/g \geq q_{1-\beta_c}$, i.e. $\varphi n + \eta \mathrm{SS} + g\, q_{1-\beta_c} \leq B$. Writing $g\, q_{1-\beta_c} = (q_{1-\beta_c}/N\mu)\,[\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)]$ gives (18). Under normality $q_{1-\beta_c} = N\mu + z_{\beta_c}\sigma_Q$, so $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$. □

The chance constraint therefore leaves the structure intact: (18) is the four-piece carbon function with the inbound and last-mile carbon rates scaled, $\varepsilon_0 \mapsto m\varepsilon_0$ and $\varepsilon_L \mapsto m\varepsilon_L$, while cost and the facility and holding terms are unchanged. Because Q is realized market throughput, exogenous to the firm's choice of n and v, the factor m is a decision-invariant scalar on the transport carbon rates; the priced objective is again (8) under this substitution, so Propositions 2–5 and Corollary 6 carry over, and only the two thresholds move.

Corollary 8 (Uncertainty tightens the thresholds)**.** *Under the chance-constrained cap the speed threshold* (11) *becomes* $\eta/h > m\,\varepsilon'(v_{\mathrm{K}})/t'(v_{\mathrm{K}})$ *(*$m\,\varepsilon_0/t_1$ *under the quadratic), and the pooling-adjusted density threshold* (15) *becomes*

$$m\,\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\, G\big(n_K(v), v\big)}{F + h\, G\big(n_K(v), v\big)}, \tag{19}$$

which reduces to $m\,\varepsilon_L/t_L > \varphi/F$ *in the no-pooling limit (*$\rho = 1$*). Since* $m \geq 1$ *for* $\beta_c \leq \frac{1}{2}$*, with equality at* $\beta_c = \frac{1}{2}$*, the chance constraint tightens the speed-reversal threshold and relaxes the dispersion threshold relative to the expected-value cap, and the effect grows as* m *rises with the demand coefficient of variation* σ/μ*, the correlation* ρ*, and the confidence* $1 - \beta_c$*.*

Proof. The chance constraint multiplies the inbound and last-mile carbon rates by m (Proposition 7); substituting $m\varepsilon_0$ and $m\varepsilon_L$ for ε_0 and ε_L in (11) and (15) gives the two conditions, the latter scaling only the left side and leaving the pooling-adjusted facility threshold $(\varphi+\eta G)/(F+hG)$ on the right. From $\sigma_Q^2 = \sigma^2 N[1 + (N-1)\rho]$, $\mathrm{CV}_Q = (\sigma/\mu)\sqrt{[1 + (N-1)\rho]/N}$ is increasing in σ/μ and, for $N > 1$, in ρ, and z_{β_c} is increasing in $1-\beta_c$, so $m = 1+z_{\beta_c}\mathrm{CV}_Q$ is increasing in all three. □

Demand uncertainty thus enters carbon twice. It enters the holding bucket through the safety stock $z\sigma\sqrt{L}$, the channel behind the speed reversal, and, under a chance-constrained cap, it enters the transport bucket through the margin $z_{\beta_c}\,\mathrm{CV}_Q$ that the firm holds against realized emissions just as it holds safety stock against demand. More variable or more correlated demand raises this margin and, like a higher service promise, shifts the carbon-minimizing configuration; the expected-value cap is the limiting case $m = 1$. Carrying the realized variability of on-hand inventory as well would place an analogous margin on the holding bucket; we keep the safety stock at its planned level, consistent with the base-stock policy. The deterministic equivalent itself is distribution-free in the throughput quantile $q_{1-\beta_c}$; the normal approximation enters only through the closed form $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$, which is most accurate for moderate N and modest CV_Q and which a firm

with strongly skewed demand can replace by the empirical throughput quantile without altering the structure.

In-transit inventory. The holding bucket is modeled through safety stock; in-transit (pipeline) inventory, $N\mu L = N\mu D/v$, falls in v and is omitted from the core. Were it carried at the warehouse holding rates, it would add $(h+\tau\eta)\,N\mu D/v$ to the holding bucket, and the speed first-order condition becomes

$$2(t_1 + \tau\varepsilon_0)N\mu D\, v \;=\; (h + \tau\eta)\Big[\tfrac{1}{2}\, z\sigma\sqrt{ND}\,\sqrt{\Psi}\, v^{-3/2} + N\mu D\, v^{-2}\Big],$$

so $(h + \tau\eta)/(t_1 + \tau\varepsilon_0)$ equals a quantity strictly increasing in v. Because this is the same ratio that governs (11), the speed threshold is *unchanged*; only the level of v^* rises (the closed form (12) is replaced by the root of this condition, no longer radical-free). If instead in-transit stock carries its capital cost but negligible warehouse carbon, the cost-minimizing speed rises while the carbon-minimizing speed does not, so (11) becomes marginally harder to satisfy. In either case omission affects the level of v^*, not the threshold that governs the reversal.

Load-factor sensitivity. The core treats per-unit transport carbon as independent of shipment size, so consolidation in this model lowers carbon only through inventory pooling and facility energy, never through transport — which it lengthens. A load-sensitive last mile would let the per-unit emission rate fall with vehicle fill, $e(q) = a/q + b$ (fuller vehicles emit less per unit), restoring a transport benefit of consolidation absent here (Bektaş and Laporte, 2011). Modeling that benefit fully would make shipment size — routing and consolidation — a third decision, which we keep outside the core to preserve the two-lever structure. We instead test the sign and magnitude of the force with a reduced-form last-mile rate that varies with density: the robustness analysis of Section 5 (Table 11) shows that a load-sensitive last mile softens the decentralization threshold (16) and, when the load effect is strong, can reverse it toward consolidation.

5 Computational Experiments

This section reports computational experiments that quantify and stress-test the analytical results in a calibrated representative instance: the dollar magnitudes are chosen so that the cost-minimizing network is interior, and the four carbon-intensity-per-dollar ratios are set to values representative of real logistics activities — line-haul and last-mile transport, refrigerated and ambient storage, and facility energy. All reported quantities are computed from the cost and carbon functions of Sections 3–4 by a finite grid search over the two decision variables n and v, so the experiments also serve as a check on the propositions.

5.1 Calibration

Table 2 lists the baseline parameters. The four ratios give $R_{\text{speed}} = (\eta/h)/(\varepsilon_0/t_1) = 0.375$ and $R_{\text{density}} = (\varepsilon_L/t_L)/(\varphi/F) = 0.75$, placing the baseline in the conventional regime where both

Table 2: Baseline calibration. Dollar magnitudes place the cost optimum in the interior; the four carbon-intensity-per-dollar ratios are set to values representative of real logistics activities.

Group	Values
Demand, geography	$N = 40$, $\mu = 20{,}000$, $\sigma = 8{,}000$ (CV 0.40), $\rho = 0.30$, $\alpha = 0.95$ ($z = 1.65$), $\delta = 120$, $D = 1$, $v \in [0.3, 3]$
Cost rates (\$)	$F = 7500$, $h = 0.50$, $t_0 = 1.00$, $t_1 = 0.05$, $t_L = 0.005$
Carbon rates (kg)	$\varphi = 3000$, $\eta = 0.15$, $\varepsilon_0 = 0.04$, $\varepsilon_L = 0.0015$
Ratios	$\eta/h = 0.30$, $\varepsilon_0/t_1 = 0.80$, $\varepsilon_L/t_L = 0.30$, $\varphi/F = 0.40$
Diagnostics	$R_{\text{speed}} = 0.375$, $R_{\text{density}} = 0.75$

comparisons fall short of one. Solving (7), the cost-minimizing configuration is $n^* = 7.8$ distribution centers replenished at speed $v^* = 1.03$, with weekly cost \$1.24M and carbon 160.5 t, divided between a transport bucket of 85.6 t and a holding bucket of 74.9 t. The carbon-minimizing configuration is slower and more consolidated, $n^* = 6.6$ and $v^* = 0.69$, lowering carbon by 5.1% at a cost increase of 1.2%. The movement is modest because the two buckets substitute: slowing or consolidating shifts carbon from one bucket to the other rather than removing it, so near a conventional product the net change is small. The magnitude of available abatement is regime-dependent, as the archetypes below show. The strict-convexity condition (14) compares the last-mile curvature $3e\,n^{-5/2}$ against the holding pooling-loss term $b(v)(1-\rho)^2\Psi^{-3/2}$, so it is guaranteed when the last-mile coefficient e is large relative to $b(v)(1-\rho)^2$ — a geographically spread region (large δ) or low demand variability (small σ/μ) — and weakest when pooling is strong and the region compact.[2] It holds at every configuration reported in this section — the baseline and all four archetypes — so each density optimum is the unique interior stationary point.

5.2 The four regimes

Holding the cost rates fixed and sweeping the two carbon-intensity ratios traces the analytical thresholds. Figure 1 plots the computed carbon-minimizing direction in the $(R_{\text{density}}, R_{\text{speed}})$ plane: the solid curves are the speed and density flip boundaries at $\rho = 0.30$, the dashed lines the no-pooling thresholds $R_{\text{speed}} = 1$ and $R_{\text{density}} = 1$. Without pooling the two boundaries separate exactly at $R = 1$, splitting the plane into the four regimes (set out, with their managerial readings, in Table 12 of Section 6). With pooling the speed boundary stays near $R_{\text{speed}} = 1$, while the density boundary bows to the right of $R_{\text{density}} = 1$ and tilts further right as R_{speed} rises: by the pooling-adjusted threshold (15), dispersion requires the last-mile ratio to clear a facility threshold that pooling pulls toward η/h, so a higher holding intensity raises the bar for dispersion. The computed boundary thus traces the mediant of Proposition 4.

[2] Explicitly, at a given speed v the binding case of (14) is $n = N$, where it reduces to $\delta > b(v)(1-\rho)^2/[3(t_L+\tau\varepsilon_L)\mu]$. At the baseline this requires $\delta > 82$ at the carbon optimum and $\delta > 67$ at the cost optimum, both met by $\delta = 120$; only the slowest speeds at the highest density (the non-convex corner noted after Corollary 6) require a larger region.

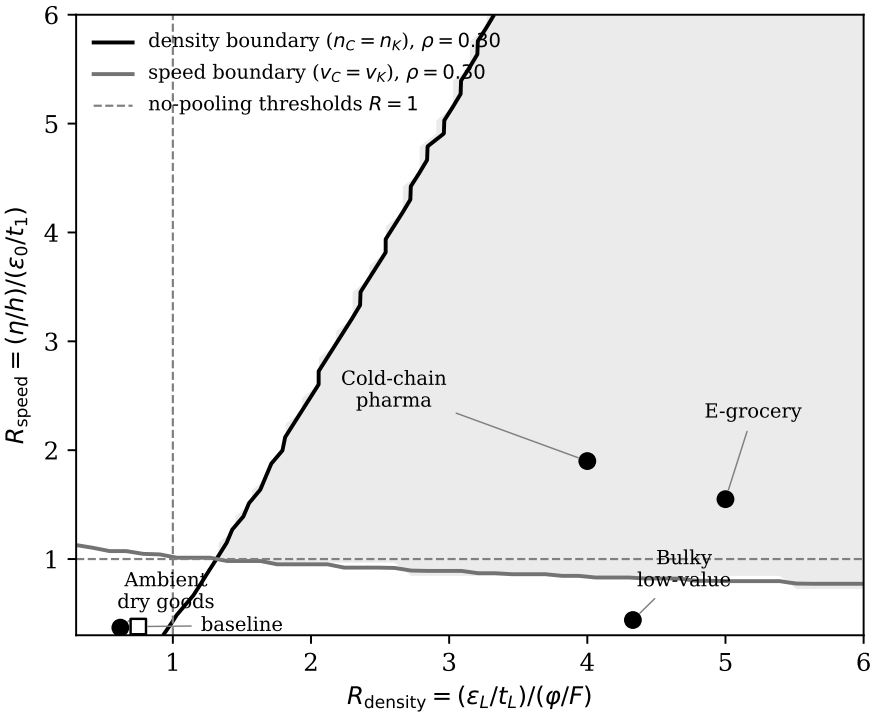


Figure 1: Computed regime boundaries in the plane of the two carbon-intensity-per-dollar ratios. Solid curves are the speed boundary ($v_C = v_K$) and the density boundary ($n_C = n_K$) at $\rho = 0.30$; dashed lines are the no-pooling thresholds $R_{\text{speed}} = 1$ and $R_{\text{density}} = 1$. The pooling-adjusted density boundary lies to the right of $R_{\text{density}} = 1$ and tilts further right as R_{speed} rises, tracing the mediant of (15). The shaded region is the faster-and-more-dispersed regime; points locate the four archetypes of Table 4 and the baseline.

5.3 Constructing the product archetypes

The four archetypes are chosen from external benchmarks rather than freely. Their two diagnostic ratios are carbon intensities per dollar, whose components are measurable, and Table 3 maps published emission factors and costs into those ratios. The emission factors are well established (national inventories and standard conversion-factor sets) and the cost figures order-of-magnitude (industry benchmarks), so the benchmarks set the *relative* magnitudes of the ratios across archetypes rather than estimate parameters.

The two strongest anchors (Table 3) are the refrigerated-warehouse energy premium, which places cold-chain and e-grocery in the high-η/h band, and the low-fill diesel last mile, which raises ε_L/t_L and underlies the load-factor result reported below. Value density enters through the holding cost, since the capital cost of inventory is part of h: a higher-value product carries a larger h and, at equal carbon, a smaller η/h. Cold-chain pharma is therefore the one archetype whose holding placement is assumption-dependent: its high value density enlarges h, so its high η/h relies on the cold-chain energy and compliance component (refrigeration, ultra-low-temperature, continuous monitoring) dominating the capital-cost denominator.

Using these anchors, we choose the normalized carbon-intensity ratios in Table 4, which span

Table 3: From published benchmarks to the model's diagnostic ratios. Emission factors are from national inventories and conversion-factor sets (high confidence); costs are industry benchmarks (order-of-magnitude). The final column lists the archetypes each anchor informs.

Empirical anchor	Observed pattern	Model implication	Archetypes
Warehouse energy, cold vs. ambient (U.S. Environmental Protection Agency, 2024, 2025)	refrigerated $\approx 3.7\times$ ambient carbon ($\approx$ 8.6 vs. 2.3 kg $CO_2e/ft^2/yr$)	raises holding carbon η, so larger η/h (favors faster replenishment)	cold-chain, e-grocery
Last-mile van vs. HGV (DESNZ/DEFRA, 2024; Smart Freight Centre, 2022)	van $\approx$ 0.62 kg CO_2e/t-km, $\approx$ 6–8$\times$ HGV, 37–41% fill	raises last-mile ε_L, so larger ε_L/t_L (favors dispersion)	e-grocery, pharma, bulky
Bulky, ambient, low-value (U.S. Geological Survey, 2025)	crushed stone $\approx$ \$17.5/tonne; minimal, inexpensive storage	low storage carbon keeps η/h low; dispersion is driven by high ε_L/t_L, not η/h	bulky low-value
High value density (U.S. Federal Highway Administration and Bureau of Transportation Statistics, 2023)	pharma $\approx$ \$49,000/tonne	large h damps η/h; pharma's high η/h relies on cold-chain/compliance energy dominating h	cold-chain pharma
Line-haul mode factors (DESNZ/DEFRA, 2025)	ocean 0.016 < rail 0.028 < road 0.10–0.21 kg CO_2e/t-km	orders inbound carbon against speed; ε_0/t_1 not identified	inbound (all)

Table 4: Carbon-intensity-per-dollar ratios for the four archetypes — representative calibrations for the four regimes — and the diagnostics $R_{\text{speed}} = (\eta/h)/(\varepsilon_0/t_1)$ and $R_{\text{density}} = (\varepsilon_L/t_L)/(\varphi/F)$ they imply.

Archetype	η/h	ε_0/t_1	ε_L/t_L	φ/F	R_{speed}	R_{density}
Ambient dry goods	0.30	0.80	0.25	0.40	0.375	0.625
Cold-chain pharma	0.95	0.50	1.40	0.35	1.90	4.00
Bulky low-value	0.20	0.45	1.30	0.30	0.44	4.33
E-grocery perishable	0.85	0.55	1.50	0.30	1.55	5.00

the four theoretical regimes; these values are regime-representative calibrations consistent with the relative magnitudes above. The grounding is order-of-magnitude: the emission factors (the numerators) are well established, but the per-dollar ratios also need cost denominators that public data pin only loosely — last-mile cost per parcel and the inbound speed coefficients ε_0, t_1 are the least directly identified from public benchmarks — which is why the experiments use these representative calibrations rather than estimated values.

Table 5: Carbon-minimizing outcomes for the four archetypes calibrated in Table 4: the direction relative to the cost optimum, the largest attainable carbon reduction ("Max cut"), and the cost premium to reach 80% of that reduction and to reach the carbon minimum.

Archetype	Carbon-min direction	Max cut	Cost @ 80%	Cost @ min
Ambient dry goods	slower, consolidate	6.1%	+0.5%	+1.5%
Cold-chain pharma	faster, disperse	7.7%	+1.1%	+3.8%
Bulky low-value	slower, disperse	17.4%	+2.2%	+7.8%
E-grocery perishable	faster, disperse	9.6%	+1.4%	+4.8%

5.4 Product archetypes

Table 5 reports the carbon-minimizing configuration for the four archetypes constructed above. Because they share the cost rates, all four have the same cost-minimizing configuration ($n^* = 7.8$, $v^* = 1.03$) and differ only in the direction and magnitude by which the carbon-minimizing configuration departs from it. Slowing and consolidating is carbon-minimizing for the ambient dry-goods archetype; the cold-chain and e-grocery archetypes call for faster and more dispersed networks; and the bulky, low-value archetype, with inexpensive storage but a carbon-intensive last mile, disperses while slowing. Available abatement ranges from about 6% to 17% of total carbon across the four; reaching 80% of that abatement costs between 0.5% and 2.2%, while reaching the carbon minimum costs between 1.5% and 7.8%, the larger premiums attaching to the products that disperse most. All four archetypes have interior density optima ($J_n(1) < 0 < J_n(N)$), so none is a corner solution.

5.5 Value of joint redesign

Because the two levers are complements (Proposition 5), adjusting them together reaches abatement and cost levels that neither reaches alone. Table 6 reports this for a product in the faster-and-more-dispersed regime (e-grocery perishables, $R_{\text{speed}} = 1.55$, $R_{\text{density}} = 5.0$). Holding the network fixed and varying only speed lowers carbon by at most 0.5%; holding speed fixed and varying only density reaches 8.4%; adjusting both reaches 9.6%. We report two targets: a *moderate* target of 7.6%, set at 90% of the best single lever's attainable abatement (here density, $0.90 \times 8.4\%$), and a *deep* target of 9.1%, set at 95% of the joint attainable abatement ($0.95 \times 9.6\%$). The moderate target is infeasible on the speed lever alone and costs 1.8% on the density lever alone, but only 1.3% when both are adjusted; the deep target lies beyond either single lever and is met only by adjusting both. Faster replenishment lowers the inventory penalty of dispersing, so the joint redesign extends the reach of each lever.

5.6 Carbon-price path

Figure 2 traces the cost–carbon frontier as the carbon price τ rises from zero, for the baseline product. The configuration moves from the cost-minimizing point to the carbon-minimizing one, with both levers moving inward together — the network consolidates from 7.8 to 6.6 DCs and

Table 6: Value of adjusting both levers, for the e-grocery archetype. "Max cut" is the largest carbon reduction the policy attains; the last two columns give the cost increase to meet a moderate and a deep carbon target.

Policy	Max carbon cut	Cost at 7.6% target	Cost at 9.1% target
Speed only (n fixed)	0.5%	infeasible	infeasible
Density only (v fixed)	8.4%	+1.8%	infeasible
Joint (n, v)	9.6%	+1.3%	+2.8%

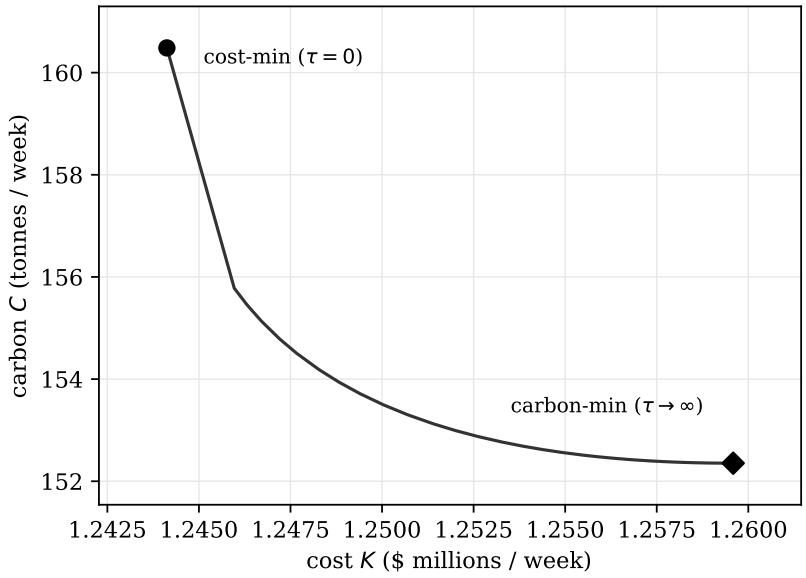


Figure 2: Cost–carbon frontier for the baseline product, tracing the cost–carbon Pareto frontier as the carbon price τ increases from 0 to ∞. The endpoints are the cost-minimizing ($\tau = 0$) and carbon-minimizing ($\tau \to \infty$) configurations.

the speed falls from 1.03 to 0.69 — and most of the movement occurs at low prices, the frontier flattening as the configuration approaches the carbon-minimizing one.

5.7 Chance-constrained carbon caps

Moving from an expected carbon cap to a cap on *realized* carbon held with high confidence is not merely a tightening of the same constraint; by Proposition 7 it scales the transport bucket by $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$ and so reweights transport emissions relative to inventory and facilities, changing the relative attractiveness of speed and density. We trace this for the baseline product. With $N = 40$, $\mu = 20{,}000$, $\sigma = 8{,}000$ (CV 0.40), and $\rho = 0.30$, the aggregate throughput coefficient of variation is

$$\mathrm{CV}_Q = \frac{\sigma}{\mu}\sqrt{\frac{1 + (N-1)\rho}{N}} \approx 0.225,$$

so the transport scalar is $m \approx 1.29$, 1.37, and 1.52 at the 90%, 95%, and 99% confidence levels ($z_{\beta_c} \approx 1.28$, 1.64, 2.33), against $m = 1$ for the expected cap.

Table 7: Effect of chance-constrained carbon accounting on the carbon-minimizing design (baseline product; cost optimum $n_K = 7.8$, $v_K = 1.03$; reference thresholds $\eta/h = 0.30$ and the no-pooling facility ratio $\varphi/F = 0.40$, against which the pooling-adjusted density threshold at the cost optimum is ≈ 0.37). "Speed" and "Network" give the direction relative to the cost optimum. Higher confidence scales the transport bucket by m, lowering the carbon-minimizing speed and, beyond the 90% level, dispersing the network.

Cap / conf.	m	$m\varepsilon_0/t_1$	$m\varepsilon_L/t_L$	v_C	n_C	Speed	Network	Cost prem.
Expected (m=1)	1.00	0.80	0.30	0.69	6.6	slower	consolidate	+1.2%
90%	1.29	1.03	0.39	0.63	7.7	slower	consolidate	+1.7%
95%	1.37	1.10	0.41	0.62	8.0	slower	disperse	+1.9%
99%	1.52	1.22	0.46	0.59	8.6	slower	disperse	+2.2%

Table 7 reports the carbon-minimizing design under the scaled carbon function at each confidence level, with ε_0 and ε_L replaced by $m\varepsilon_0$ and $m\varepsilon_L$. Two effects appear, consistent with Corollary 8. On speed, the effective inbound ratio $m\varepsilon_0/t_1$ rises from 0.80 to 1.22, moving further above the holding ratio $\eta/h = 0.30$, so the carbon-minimizing speed falls at every confidence level, from 0.69 to 0.59. On density, the effective last-mile ratio $m\varepsilon_L/t_L$ rises from 0.30 to 0.46, raising the risk-weighted last-mile term against the pooling-adjusted facility threshold of Proposition 4. That proposition compares density best responses at a *fixed* speed; the table instead reports the *joint* carbon-minimizing density n_C against the joint cost-minimizing density n_K, in which the speed also adjusts. On that joint basis the network shifts from more consolidated than the cost optimum (at the expected and 90% caps) to more dispersed (at the 95% and 99% caps), so the crossing of n_K occurs between the 90% and 95% levels. For this product a high-confidence carbon cap therefore calls for a chain that is slower and progressively less consolidated, and beyond a threshold confidence more dispersed than the cost optimum — a change in direction produced by the accounting rule alone, with the physical parameters held fixed. Figure 3 traces the two designs across confidence levels.

A high-confidence carbon cap changes the design calculation by placing a risk margin on transport emissions: it makes faster replenishment less attractive, because inbound transport emissions receive a higher effective weight, and it makes dispersion more attractive, because last-mile emissions receive a higher effective weight. Two comparisons must be kept apart here. At a *fixed* speed, Proposition 4 disperses the density best response past the cost optimum once the risk-scaled last-mile ratio $m\,\varepsilon_L/t_L$ clears the pooling-adjusted threshold (≈ 0.37 at the cost optimum, below the no-pooling divider $\varphi/F = 0.40$); by that fixed-speed test dispersion would already set in at the 90% cap, where $m\,\varepsilon_L/t_L = 0.39$. The table and Figure 3, however, report the *joint* carbon-minimizing density n_C against the joint cost optimum n_K, and the risk margin also lowers the speed; by complementarity (Proposition 5) a slower chain calls for fewer warehouses, which partly offsets the dispersion. The net effect is that the joint n_C rises through n_K only between the 90% and 95% levels — later than the fixed-speed crossing. The fixed-speed threshold and the joint crossing are thus distinct and should not be equated. Demand uncertainty enters the carbon design through

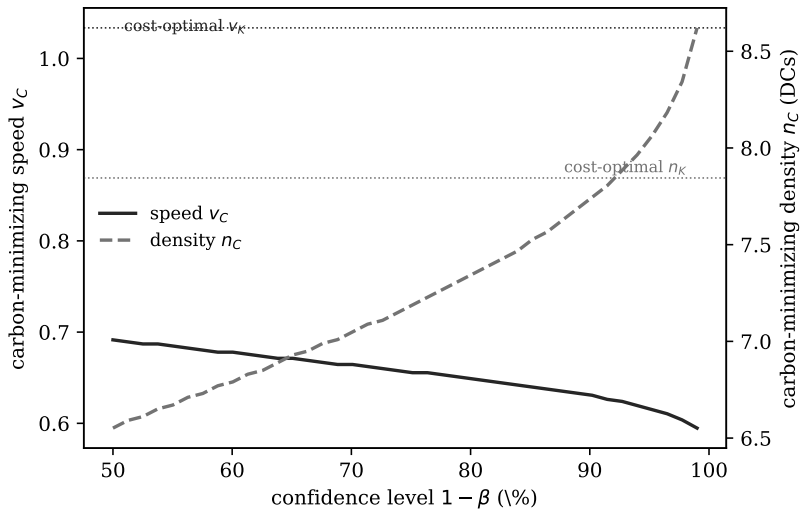


Figure 3: Carbon-minimizing speed v_C (left axis, solid) and density n_C (right axis, dashed) under increasingly stringent confidence levels for the baseline product. Dotted lines mark the cost-optimal speed v_K and density n_K; the carbon-minimizing speed stays below v_K throughout, while the carbon-minimizing density rises through n_K between the 90% and 95% confidence levels.

Table 8: Chance-constrained cap at 95% confidence across demand variability. Higher demand CV raises the throughput-risk scalar m; the network direction is stated relative to the (CV-dependent) cost optimum.

Demand CV	CV_Q	cost opt. (n_K, v_K)	m at 95%	carbon-min (n_C, v_C)	Network vs. cost
0.20	0.113	(8.7, 0.79)	1.19	(8.0, 0.50)	consolidate
0.40	0.225	(7.8, 1.03)	1.37	(8.0, 0.62)	disperse
0.60	0.338	(7.2, 1.21)	1.56	(8.1, 0.69)	disperse

two channels: safety stock on the inventory side and a throughput-risk margin on the transport side.

Sensitivity to demand variability. The transport scalar $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$ rises with the demand coefficient of variation, since $\mathrm{CV}_Q = (\sigma/\mu)\sqrt{[1 + (N-1)\rho]/N}$ is increasing in σ/μ. Table 8 repeats the 95% cap at three demand CVs. A more variable demand raises m — from 1.19 at $\mathrm{CV} = 0.20$ to 1.56 at $\mathrm{CV} = 0.60$ — and so deepens the transport reweighting and the reduction in carbon-minimizing speed. The carbon-minimizing density at the 95% cap is roughly flat near eight DCs, while the cost optimum consolidates as variability grows (more safety stock raises the value of pooling), so relative to the cost optimum the high-confidence network is more consolidated at low CV and more dispersed at high CV.

Monte-Carlo validation. Because the chance constraint has a deterministic equivalent, no simulation is needed to solve it; a Monte-Carlo check nonetheless confirms that the normal approximation delivers the intended coverage. Fixing the design at the cost optimum and setting the cap

Table 9: Monte-Carlo validation of the chance constraint (5×10^5 draws; design fixed at the cost optimum, cap at the normal deterministic equivalent). Empirical violation frequency $\Pr\{\tilde{C} > B\}$ under normal and right-skewed (gamma) throughput of equal mean and variance.

	Nominal β_c	Normal demand	Gamma demand
Demand CV $= 0.40$ *(*$\mathrm{CV}_Q = 0.225$*)*			
90% cap	0.100	0.100	0.105
95% cap	0.050	0.050	0.061
99% cap	0.010	0.010	0.019
Demand CV $= 0.80$ *(*$\mathrm{CV}_Q = 0.451$*)*			
90% cap	0.100	0.100	0.108
95% cap	0.050	0.050	0.068
99% cap	0.010	0.010	0.026

at the normal deterministic equivalent for each confidence level, we draw 5×10^5 realizations of aggregate throughput and record the empirical frequency with which realized carbon exceeds the cap (Table 9). Under normally distributed throughput the empirical violation rate matches the nominal β_c to within sampling error, confirming the algebra of Proposition 7. Under right-skewed (gamma) throughput of equal mean and variance the violation rate exceeds the nominal level, and the gap widens at higher confidence and higher CV — the 99% cap admits a 1.9% violation at CV = 0.40 and 2.6% at CV = 0.80 — so a firm facing strongly skewed demand should set z_{β_c} from the empirical throughput quantile rather than the normal one, as the distribution-free form of Proposition 7 permits.

5.8 Robustness

The robustness analysis comprises four checks. Table 10 reports the first three. Restricting the network to an integer number of DCs returns the continuous result rounded (cost eight, carbon seven). Restricting replenishment to three discrete modes ($v \in \{0.5, 1.0, 2.0\}$) leaves the cost optimum at the standard mode and moves the carbon optimum to the slow mode, the discrete counterpart of the continuous speed reduction. Varying the demand correlation ρ shifts both optima toward more facilities as pooling weakens, with the carbon optimum consolidated relative to the cost optimum throughout. The fourth check, reported separately in Table 11, allows the last-mile emission rate to fall with vehicle fill — a load factor $\varepsilon_L\,(b + a\,n/n_{\mathrm{ref}})$ that is half load-sensitive at the reference density — adding a transport benefit of consolidation: it lowers the carbon-minimizing density sharply, from 6.6 to 3.0 DCs for the baseline and from 22.8 to 5.8 for the bulky archetype. The dispersion conclusions are therefore conditional on shipment-size-independent transport rates; load-sensitive rates shift the boundary toward consolidation.

Table 10: Robustness of the baseline cost- and carbon-minimizing designs to integer density, discrete speed modes, and demand correlation. Entries are (n, v) except the correlation rows, which report density n.

Variation	Cost-min	Carbon-min
Continuous baseline	(7.8, 1.03)	(6.6, 0.69)
Integer n	(8, 1.04)	(7, 0.69)
Discrete modes $v \in \{0.5, 1, 2\}$	(7.8, 1.0)	(6.3, 0.5)
Correlation $\rho = 0.0$	$n = 5.9$	$n = 4.8$
Correlation $\rho = 0.3$	$n = 7.8$	$n = 6.6$
Correlation $\rho = 0.6$	$n = 8.9$	$n = 7.4$
Correlation $\rho = 0.9$	$n = 9.8$	$n = 8.1$

Table 11: Load-factor sensitivity. With a last mile that is half load-sensitive at the reference density, $\varepsilon_L\,(b + a\,n/n_{\text{ref}})$, fuller vehicles under consolidation lower the carbon-minimizing density; for the bulky archetype this reverses the dispersion of Table 5.

	Carbon-min n, constant rate	Carbon-min n, load-sensitive
Baseline product	6.6	3.0
Bulky low-value archetype	22.8	5.8

6 Managerial Implications

The carbon-minimizing configuration of speed and density depends on the two carbon-intensity-per-dollar comparisons established in Section 4: holding against inbound transport, which governs speed, and last-mile delivery against facilities, which governs density. Table 12 arranges the four combinations; slowing and consolidating — the common transport levers — correspond to the lower-left cell, and the carbon-minimizing direction differs in the other three. The split on density uses the facility comparison; under pooling the dividing line shifts toward the holding comparison (Proposition 4), so a strongly pooled network needs a more carbon-intensive last mile to justify dispersing.

Diagnosis. The two ratios, computed for a given product, locate it in one cell of Table 12, which indicates the carbon-minimizing direction of speed and density. A uniform policy of slowing replenishment and consolidating the network coincides with the carbon-minimizing configuration only for products in the lower-left cell.

A decision procedure. The diagnosis suggests a concrete sequence for a firm holding a carbon target. *(i)* From its own rate data, estimate each activity's carbon intensity per dollar — for holding, inbound transport, last-mile delivery, and facilities. *(ii)* For speed, compare holding with inbound transport: if holding is the more carbon-intensive per dollar, the carbon-minimizing direction is faster replenishment, and otherwise slower (Proposition 2). *(iii)* For density, compare last-mile delivery with the pooling-adjusted facility threshold of Proposition 4 rather than with facilities

Table 12: Direction the carbon-minimizing configuration moves relative to the cost optimum, by the two carbon-intensity-per-dollar comparisons. The density columns split at the facility comparison; under risk pooling the dividing line is the pooling-adjusted threshold of Proposition 4, which lies between the facility and holding comparisons, so the facility comparison is a reference, not the exact boundary.

	Lower network density *last mile less carbon-intensive per dollar than facilities*	**Higher network density** *last mile more carbon-intensive per dollar than facilities*
Higher replenishment speed *holding more carbon-intensive per dollar than inbound transport*	faster and more consolidated, pooling inventory upstream — carbon-intensive inventory with a lower-emission or short last mile	*faster and more dispersed* — cold-chain, volatile, high-service goods on a diesel last mile
Lower replenishment speed *holding less carbon-intensive per dollar than inbound transport*	slower and more consolidated — the transport-lever case; ambient, stable, low-service dry goods	slower long haul, denser network — lower-emission, low-cost storage with carbon-intensive inbound and last mile

alone — the more strongly the network pools, the more carbon-intensive the last mile must be to justify dispersing. *(iv)* These comparisons give the carbon-minimizing *direction* on each axis, not a single-lever implementation rule: because the levers are complements, set the two together rather than in sequence, since the chosen speed raises or lowers the value of dispersing, and conversely. *(v)* If the cap is on realized rather than expected carbon, inflate the transport intensities by the throughput risk margin (Proposition 7) before applying steps (ii)–(iii).

Replenishment speed. For refrigerated, volatile, or high-service products — those for which inventory is the emissions-intensive component — a tighter carbon target is met by shipping *faster*, because the safety stock that a slower mode would require carries more carbon than the faster transport adds. This case corresponds to the top row of Table 12.

Consolidation and the last mile. Consolidation lowers carbon here through inventory pooling and reduced facility energy while lengthening the last mile. A firm whose last mile is carbon-intensive — diesel vans, long rural routes — can lower total carbon by holding more stock nearer to demand, even though that raises total inventory; the transport channel of consolidation works against it. This last channel is strong when vehicle fill improves with consolidation: under a load-sensitive last mile the carbon-minimizing network can swing sharply back toward consolidation — for the bulky archetype, from 22.8 to 5.8 DCs (Section 5) — so a firm should weigh its last mile's load sensitivity before dispersing.

Joint adjustment of the two levers. Because the levers are complements, a program that adjusts only one — a slow-steaming mandate or a network-consolidation initiative — attains part of the available abatement, and for some products moves a lever away from its carbon-minimizing direction. The cost of working one lever at a time is quantifiable: for the e-grocery archetype (Section 5, Table 6), adjusting speed alone abates at most 0.5% and density alone 8.4%, while adjusting both reaches 9.6%, and a target of 7.6% that costs 1.8% on the density lever alone costs only 1.3% when both move. Joint redesign can thus expand the abatement reach and, in this illustration, lowers the cost of meeting the target relative to single-lever adjustment, which supports evaluating the two levers together.

Product segmentation, service, and volatility. Because the carbon-minimizing configuration depends on the product, a single chain-wide policy serves a heterogeneous portfolio imperfectly. The service-level target and demand variability also bear on carbon: a higher service-level target and a more volatile demand both enlarge safety stock and so, by Proposition 3, raise the carbon-minimizing speed and strengthen the pull toward consolidation through the value of pooling, unless the last-mile comparison dominates the density decision. A 99% cycle-service target is thus a carbon-design parameter, not only an inventory-cost one. Higher demand correlation across markets weakens the pooling benefit that favors consolidation and can make dispersion more attractive, so forecasting and demand-smoothing are carbon levers in their own right.

Carbon accounting and confidence. Whether the cap is placed on expected or on realized carbon is itself a design parameter. A cap on realized carbon held with high confidence places a risk margin on transport emissions (Proposition 7), lowering the attractiveness of faster replenishment and raising that of dispersion. For the baseline product, raising the confidence level lowers the carbon-minimizing speed at every level and, beyond a threshold confidence, shifts the network from consolidation to dispersion relative to the cost optimum (Section 5, Table 7); whether that shift occurs depends on whether the risk-weighted last-mile term overcomes the facility and pooling forces.

Implication for policy. The two-bucket structure carries a caution for regulation. A rule that prices or caps only transport leaves the holding bucket untouched and can induce substitution into inventory and facilities to comply, raising emissions the rule does not measure. Instruments scoped to transport alone — a freight fuel levy, a mode-shift mandate, or a vehicle-emissions standard — are exposed to this leakage, whereas an instrument that prices a firm's total logistics carbon (for example, a scope-inclusive carbon charge or a cap covering both warehousing and transport emissions) avoids it and lets the firm choose the speed-and-density mix that minimizes carbon across transport and inventory together.

7 Discussion and Conclusion

A distribution network provides two services — availability and proximity — each of which can be supplied through more than one operational substitute, and carbon-aware design turns on which substitute a carbon price makes cheaper in emissions terms rather than on a single transport-centered response. The appropriate configuration therefore depends on how a carbon price changes the relative cost of providing those two services. When inventory is the emissions-intensive way to provide availability, carbon favors buying it with replenishment speed rather than with safety stock; when the last mile is the emissions-intensive way to provide proximity, carbon favors buying it with facilities rather than with distance. The contribution of the paper is to make these substitutions explicit and to give the thresholds at which the carbon-minimizing substitute differs from the cost-minimizing one.

Theoretical contribution. The model embeds an endogenous replenishment speed in a location–inventory network. Speed sets the lead time, the lead time sets the safety stock, and the safety stock carries carbon, so a faster network is more carbon-intensive in transport but less inventory-intensive. This coupling produces the speed threshold $\eta/h > \varepsilon'(v_{\mathrm{K}})/t'(v_{\mathrm{K}})$ (Proposition 2): when holding is more carbon-intensive per dollar than inbound transport at the margin, the carbon-minimizing chain replenishes faster than the cost-minimizing one. The density decision adds the proximity substitution, with a *pooling-adjusted* threshold (Proposition 4) in which last-mile intensity is compared not to the facility ratio φ/F but to a weighted average of φ/F and the holding ratio η/h, so risk pooling enters the boundary explicitly. The two levers are complements (Proposition 5): faster replenishment lowers the inventory penalty of dispersion, so the optimal degree of centralization is itself contingent on speed. The chance-constrained extension (Proposition 7) preserves this structure: a cap on realized rather than expected carbon scales the transport bucket by $m = q_{1-\beta_c}/(N\mu)$, leaving the four-piece form and the comparative statics intact while shifting both thresholds.

Managerial and policy implications. For practice, the diagnostic is to measure carbon intensity per dollar by activity and to locate each product in the resulting regime (Section 6). Because the carbon-minimizing configuration depends on the product, a single chain-wide policy serves a heterogeneous portfolio imperfectly, and service-level targets and demand volatility are carbon-design parameters rather than inventory-cost parameters alone. In the e-grocery experiment, joint redesign expands abatement reach and lowers the cost of the target relative to single-lever adjustment, and a high-confidence carbon cap can alter the relative attractiveness of speed and dispersion. For policy, the two-bucket structure cautions that regulation aimed only at transport can induce substitution into inventory and facility emissions that the rule does not measure; total logistics carbon is the appropriate accounting boundary, and a chance-constrained or high-confidence target places particular weight on the variability of transport throughput.

Limitations. The model is deliberately parsimonious. Markets are symmetric and treated through a continuous-approximation geometry; the product is single; inbound distance is represented by a single representative mean; the core model carries no explicit vehicle load-factor decision; service is modeled through a cycle-service level rather than a fill rate; the expected-value cap is the baseline against which the chance-constrained cap is compared; and the calibration is regime-representative rather than an empirical estimation. Each abstraction trades realism for the closed-form structure that makes the substitutions transparent. The core also abstracts from load-sensitive transport rates; the robustness experiments show that a load-sensitive last mile can attenuate or even reverse dispersion. The dispersion conclusions are thus conditional on shipment-size-independent transport rates, and on the omission of inbound consolidation and facility scale economies; each of these forces shifts the boundary toward consolidation.

Future research. Several extensions follow naturally: multi-product portfolio design, in which a shared network serves products with different carbon-intensity profiles; explicit load-factor and shipment-consolidation emissions, which would add shipment size as a third lever; heterogeneous geography and discrete facility locations in place of the continuous approximation; empirical calibration using firm-level logistics-emissions data; endogenous service-level choice, which Proposition 3 treats as a fixed carbon parameter but which a fuller model would set jointly with speed and density, internalizing the carbon cost of a service promise; and multi-echelon safety-stock placement, which would couple the density and speed decisions across more than one stocking tier.

Carbon-minimizing distribution design requires identifying the lower-emission operational substitute for availability and for proximity. In some regimes that substitute is slower replenishment and consolidation; in others it is faster replenishment, greater dispersion, or both. The model provides a diagnostic structure — two carbon-intensity-per-dollar comparisons, coupled through the complementarity of speed and density — for making that distinction.

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